DSA Tutorial - 3

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Topic: Sorting Algorithms

INPUT :

```
#include <stdio.h>
void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}

void bs(int arr[],int n){
    for (int i = 0; i < n - 1; i++) {
        int swapped = 0;
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
                swapped = 1;
            }
        }
        if (swapped == 0)
            break;
    }

    printf("Sorted array using Bubble
sort: ");
    for (int i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");
}

void ss(int arr[], int n) {
    for (int i = 0; i < n - 1; i++) {
        int minidx = i;
        for (int j = i + 1; j < n; j++) {
            if (arr[j] < arr[minidx]) {
                minidx = j;
            }
        }
        if (minidx != i) {
            int temp = arr[i];
            arr[i] = arr[minidx];
            arr[minidx] = temp;
        }
    }
    printf("Sorted array using selection
sort: ");
    for (int i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");
}

void is(int arr[], int n){
    for(int i=1;i<n;i++){
        int key = arr[i];
        int j = i-1;
        while (j>=0 && arr[j]>key){
            arr[j+1] = arr[j];
            j--;
            arr[j+1]= key;
        }
```

```
    }
    printf("Sorted array using Insertion
sort: ");
    for (int i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");
}

int partition(int arr[],int low,int high){
    int pivot = arr[high];
    int i = low -1 ;
    for (int j = low;j<high;j++){
        if(arr[j]<pivot){
            i++;
            swap(&arr[i],&arr[j]);
        }
    }
    swap(&arr[i+1],&arr[high]);
    return (i+1);
}

void qs(int arr[],int l, int high){
    if(l<high){
        int pi = partition(arr,l,high);
        qs(arr,l,pi-1);
        qs(arr,pi+1,high);
    }
}

void merge(int arr[], int l, int m, int h,
int sa[]) {
    int i = l;
    int j = m + 1;
    int k = l;

    while (i <= m && j <= h) {
        if (arr[i] <= arr[j]) {
            sa[k] = arr[i];
            i++;
        } else {
            sa[k] = arr[j];
            j++;
        }
        k++;
    }

    while (i <= m) {
        sa[k] = arr[i];
        i++;
        k++;
    }

    while (j <= h) {
        sa[k] = arr[j];
        j++;
        k++;
    }

    for (int x = l; x <= h; x++) {
        arr[x] = sa[x];
    }
}

void ms(int arr[], int l, int h, int sa[]) {
    if (l < h) {
        int m = l + (h - l) / 2;
        ms(arr, l, m, sa);
        ms(arr, m + 1, h, sa);
        merge(arr, l, m, h, sa);
    }
}
```

```
int main() {
  int n;
  int c;
  printf("Enter Number of elements:
");
  scanf("%d",&n);
  int arr[n];
  printf("Viraj T B25ET1053");
   printf("\nEnter Number of
  elements: ");
  for(int i=0;i<n;i++){
      scanf("%d",&arr[i]);
  }
 int sa[n];
printf("\nChose method: \n1. Bubble
Sort\n2. Selection sort\n3. Insertion
sort\n4. Quick Sort\n5. Merge
Sort\n\n");
 scanf("%d",&c);
switch (c) {
    case 1:
        bs(arr,n);
        break;
    case 2:
        ss(arr,n);
        break;
    case 3:
        is(arr,n);
        break;
    case 4:
        qs(arr,0,n-1);
            printf("Sorted array using
Quick sort: ");
    for (int i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");
        break;

    case 5:
        ms(arr,0,n-1,sa);
        printf("Sorted array using Merge
sort: ");
    for (int v = 0; v < n; v++) {
        printf("%d ", arr[v]);
    }
    printf("\n");
        break;

    return 0;
}
}
```

OUTPUT :

